THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the electrolysis of water, 20 mL of oxygen is collected. The volume of hydrogen collected is: 1

(a) 40 mL **(b)** 10 mL **(c)** 20 mL **(d)** 60 mL

2. An iron nail is kept in CuSO4 solution for a while. Its mass: 1

(a) remains same **(b)** increases slightly

(c) becomes zero **(d)** decreases

3. Assertion (A): Zinc displaces copper from copper sulphate solution. Reason (R): Zinc is more reactive than copper. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

4. The total number of atoms on the product side of $2\,Al + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3\,H_2$ is: 1

(a) 26 **(b)** 21 **(c)** 17 **(d)** 23

5. A gas burns with a pop sound with a lighted splinter. The gas is: 1

(a) oxygen **(b)** carbon dioxide **(c)** nitrogen **(d)** hydrogen

6. Assertion (A): Silver chloride is stored in dark bottles. Reason (R): It decomposes in the presence of sunlight. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

7. The sum of coefficients in the balanced equation $Ca(OH)_2 + H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + H_2O$ is: 1

(a) 10 **(b)** 14 **(c)** 9 **(d)** 12

8. Iron nails kept in (i) dry air, (ii) boiled water with oil layer and (iii) tap water. Rust forms in: 1

(a) all three **(b)** tubes (i) and (ii)

(c) tube (iii) only **(d)** tube (i) only

9. **Assertion (A):** Rusting of iron is an oxidation process. 1

Reason (R): During rusting, iron is reduced by oxygen and moisture.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Decomposition of calcium carbonate on heating is an exothermic reaction. 1
Reason (R): Decomposition reactions require energy in the form of heat, light or electricity.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Identify the substance oxidised and reduced in $H_2S + Cl_2 \longrightarrow 2\,HCl + S$. 2

12. Write the equation for quick lime reacting with water, and the reaction of the product with carbon dioxide. 2

13. Write the balanced equation with state symbols for the reaction: aqueous sodium sulphate + aqueous barium chloride gives solid barium sulphate + aqueous sodium chloride. 2

14. Why do silver articles turn black in air? Name the black compound. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance $Mg + SO_2 \longrightarrow MgO + S$ and say which substance is reduced. 3

16. Substance X, obtained by heating limestone, reacts vigorously with water. Identify X and the product Y, giving both equations. 3

17. Silver nitrate solution is added to sodium chloride solution. State the observation, equation and type. 3

OR

6.5 g of zinc reacts with excess dilute HCl. Using Zn = 65 and H = 1, find the mass of hydrogen liberated. 3

18. Butter kept in a refrigerator stays fresh longer than at room temperature. Explain, and suggest two other methods. 3

OR

Show that the combustion of magnesium is both a combination and a redox reaction. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meena placed a clean iron nail in a blue copper sulphate solution. After about 30 minutes she observed that the nail was covered with a brownish deposit and the blue colour of the solution had changed to pale green. 5
(i) What is the brown deposit on the nail?
(ii) Name the type of reaction.
(iii) Write the balanced equation.
(iv) Meena now dips a copper strip in iron(II) sulphate solution. Predict, with reason, what she will observe.

20. 100 kg of limestone is heated (Ca = 40, C = 12, O = 16). Write the equation, find the mass of quick lime and of carbon dioxide, name the type and give one use of quick lime. 5

OR

Lead nitrate and potassium iodide solutions are taken in a flask and in a tube kept inside the flask. Describe the set-up, observation and inference about mass. 5

END OF QUESTION PAPER — **Total: 40 Marks**